

COLOUR KEY:

Welcome — Anirban

Keynote — Ichiro

Facilities — John

TEM context — Lourdes

Programming — Erik

FFT — Alex & John

Open science — Eva

Docker — John & Karan

Break / admin

MORNING

8:00 AM	Breakfast Provided by organizers	
8:30 AM	Arrival & environment check Laptops online · Jupyter running · data in place	10 min
8:40 AM	Workshop Welcome — Anirban Mandal What is CITEAM · workshop goals · what to expect today	10 min
8:50 AM	Keynote — Prof. Ichiro Takeuchi 20 min talk + 5 min Q&A · Open science & data-driven discovery in materials science	25 min
9:15 AM	The UMD TEM facility — John Cummings JEM-ARM200F capabilities · facility access · how to get training	20 min
9:35 AM	S 0.1.1 Introduction to TEM — Lourdes Salamanca ADF detector · Z-contrast · SiGe & TbFeO ₃ example datasets	20 min
9:55 AM	S 0.1.1 First Steps (programming) — Erik Scott Imports · open TIFF · image stats · thumbnail · HyperSpy reload · metadata	25 min
10:20 AM	Break	15 min
10:35 AM	S 0.1.2 Histograms & contrast — Lourdes Detector response · dynamic range · saturated pixels on ADF	5 min
10:40 AM	S 0.1.2 Histograms & contrast (programming) — Erik plot_histograms · binning strategies · percentile clipping · gamma correction	10 min
10:50 AM	S 0.1.3 Noise correction — Lourdes Noise sources in STEM · when denoising is and is not appropriate	5 min
10:55 AM	S 0.1.3 Gaussian blur & HyperSpy denoising — Erik scipy Gaussian filter · HyperSpy decomposition · before/after comparison	10 min
11:05 AM	S 0.1.4 Drift correction — Lourdes Drift vs. jitter · why frames never align · physics of thermal drift	10 min
11:15 AM	S 0.1.4 Drift correction on a stack (programming) — Erik Load stack · estimate_shift2D · align · sum aligned · axes scale · sidecar metadata	20 min

LUNCH

11:35 AM	Lunch Provided by organizers	55 min
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AFTERNOON

12:30 PM	Microscopy suite visit Site visit to the JEM-ARM200F · led by John Cummings	60 min
1:30 PM	Open science & data management — Eva Campo What is CITEAM · FAIR data · reproducible workflows · (flexible slot — adjusts for late returns)	30 min
2:00 PM	S 0.1.5 Measuring distances — Lourdes Lattice spacings · how the scale bar gets set · calibration precision	10 min
2:10 PM	S 0.1.5 Distance measurement (programming) — Erik axes_manager GUI · sidecar .txt scale · interactive calibrate() tool	10 min
2:20 PM	Break	15 min
2:35 PM	S 0.1.6 FFT analysis — Lourdes Reciprocal space · lattice to FFT spots · masking & inverse FFT conceptually	10 min
2:45 PM	S 0.1.6 FFT analysis (programming) — Alex Hunt & John Cummings 1D line profile · rectangular ROI to FFT · polygon mask · inverse FFT · error cell walkthrough	60 min
3:45 PM	Docker install on personal laptops John Cummings (Windows / PCs) · Karan Vahi (Macs) · parallel tracks	30 min
4:15 PM	Final Q&A & close Open questions · office hours announcement · next steps	15 min
4:30 PM	End	

SESSION TOTALS				
10 min Welcome — Anirban Mandal	25 min Keynote — Ichiro Takeuchi	20 min Facilities talk — John	58 min Lourdes TEM sections total	75 min Erik programming (est.)
60 min Alex & John FFT	30 min Eva — open science	30 min Docker install	45 min Breaks	115 min Lunch + microscopy visit